

Recorded-salutation predictability from wearable epoch data

A synthesis of the September 2026 experiments · Document date 22 September 2026 · Reporting only — no model was refit for this document.

Inputs: frozen artifacts in `experiments/*/results/`, `artifacts/`, `artifacts_sq/`; per-experiment reports cited in Table 1. Build script: `reports/build_meta_report_2026-09-22.py`.

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1 Data and label

The target is the participant's *recorded* salutation field in the Datenspende cohort export, taking the two populated codes `10` and `20`. Across the 3,848-participant Garmin cohort that anchors the newest experiments, code `20` is the majority class (prevalence 0.64; class mean of `temporal_predictions.csv` = 0.642). Recorded salutation is what the participant entered; it is not established biological sex or gender, and the design records no physiological or causal claim.

The newest experiments use a single Garmin subcohort (`source_id == 3`) with the first 40 adequate days per participant (the R1b cohort). For each participant-day we have a series of heart-rate epochs with timestamps and a local timezone, plus recording-volume metadata. We do not use anthropometrics or inter-beat intervals — those fields are absent from the export.

2 Measurement protocol

For a model producing a real-valued score s_i per participant, the primary ranking metric is the area under the receiver operating characteristic curve. It equals the probability that a randomly chosen positive scores above a randomly chosen negative (with ties split):

$$\text{AUROC} = P(s^+ > s^-) + \frac{1}{2} \cdot P(s^+ = s^-) \quad (1)$$

Two designs are matched on participants and split: we report *paired* per-allocation deltas. Allocation `r` is one of $R = 10$ repeated train/validation/test partitions (allocator folds, R1a-AF). For two arms A and B we compute $\Delta_r = \text{AUROC}_{A,r} - \text{AUROC}_{B,r}$ and the 95% Bonferroni two-sided simultaneous interval:

$$\text{mean } \Delta \pm t_{1-\gamma/2, R-1} \cdot \text{SD}(\Delta) / \sqrt{R}, \quad \gamma = 0.05 / (2m), \text{ for } m \text{ primary comparisons} \quad (2)$$

Variability across allocations is model-and-split randomization conditional on the observed cohort; it is not population uncertainty. The test split is reused across the recent experiments and is therefore exploratory by predeclared rule; conclusions about absolute generalization beyond this cohort are not established.

3 Results by phase

Table 1 lists all 14 experiment stages in chronological order. Numbers are taken from each stage's committed report (cited in the rightmost column); this synthesis does not recompute them.

Table. The 14 experiment stages of the September 2026 campaign.

Stage	Cohort / design	Main result	Source
Pooled v1	20,485 all-source; default RF	test AUROC 0.7132	REPORT.md
Pooled v2	20,485 + demographics/source indicators; new exclusions	test AUROC 0.6833	artifacts_v2/model_report.md
Strict source-specific	Garmin 3,848; Apple 4,446; Samsung 541	source-specific summaries beat demographics	artifacts_sq/sidequest_report.md
RF tuning	shared config adopted	max_features=0.4, min_samples_leaf=10, class_weight=balanced_subsample, depth unrestricted, n=100	artifacts_sq/tuned_500_comparison.md
Feature selection (30 splits)	hygiene, corr prune, top-k	no consistent gain; selected subsets unstable	artifacts_sq/fsplit/feat_sel_report.md
Feature engineering	interactions, ratios, rolling diffs	max +0.0034 AUROC, intervals cross zero	artifacts_sq/feng/feat_eng_report.md
Informative recording / missingness	25,784 ungated; recording + value model	combined 0.7239; recording model adds +0.0167	2026-09-20_informative_recording_features/README.md
Continuous age vs age groups	Garmin	0.7445 → 0.7475 (+0.0029, CI [-0.0008, +0.0067])	experiments/age_vs_agegroup_garmin_2026-09-20/results/REPORT.md
OR-Tools 50/50, source-balanced	Garmin / Apple / Samsung	G 0.7290/0.7094; A 0.6875/0.6787; S 0.6289/0.6493	2026-09-21_ortools_source_50_50_salutation_rf/PLAN.md
R1a circadian features	Garmin	baseline 0.7434 → combined 0.7463 (+0.0029)	experiments/r1a_circadian_garmin_2026-09-21/results/REPORT.md
Day-scale modelling	Garmin, k days	0.6653 (k=1) → 0.7269 (k=32) → 0.7302 (all days)	experiments/dayscale_garmin_2026-09-21/results/REPORT.md
R1a-AF (R=10 allocator folds)	Garmin, 10 allocations	baseline 0.7415 → combined 0.7509 (+0.0094, allocs-positive, Bonferroni excludes o)	experiments/r1a_allocfolds_garmin_2026-09-21/results/REPORT.md
R1b date-aware circadian	Garmin, R1a-AF folds, first 40 days	day-level 0.6589 → 0.6782 (+0.0193); participant 0.7415 → 0.7513 (+0.0098)	experiments/r1b_dateaware_garmin_2026-09-21/results/REPORT.md
Temporal representations (R1b cohort/folds)	Garmin, 10 allocations, 288 × 5-min bins × 40 days	summaries 0.7031 → Combined 0.8565 test (+0.153 vs Summary_RF, Bonferroni)	experiments/temporal_garmin/results/REPORT.md

3.1 Pooled baselines and source structure

The pooled models (v1, v2) trained on all wearable sources at once showed that demographics alone are weak and that source exclusions and feature inclusion matter: the v2 protocol (added demographics and source indicators, changed exclusions) moves test AUROC from 0.7132 to 0.6833 — a difference that is not attributable to features alone because the cohort and split changed simultaneously. Source-specific models (Garmin/Apple/Samsung) showed that different sources have different useful representations.

3.2 RF tuning, feature selection, feature engineering

Tuning established the shared configuration used in every later experiment: `max_features=0.4`, `min_samples_leaf=10`, `class_weight='balanced_subsample'`, depth unrestricted, `n_estimators=100`. The two follow-up ablations tested hand-built transformations (interactions, channel ratios, rolling-window differences) and feature selection (hygiene, correlation pruning, top-k). Neither offers a reliable general improvement: feature engineering contributes at most $+0.0034$ with intervals that cross zero; hygiene reduces matrix width without a detected loss, while correlation pruning and top-k produce unstable subsets.

3.3 Recording patterns are informative

The informative-recording experiment (25,784 ungated participants) separated recording-pattern features from value features. The combined model reaches 0.7239: the recording model contributes $+0.0167$ beyond the value model. This establishes that *how much* and *when* data are recorded carries information about the recorded salutation, beyond *what* is recorded. Native NaN handling offered no detectable advantage over median imputation.

3.4 Circadian shape and day-scale

R1a added 35 hourly-profile and cosinor features; the increment is small (baseline 0.7434 $\rightarrow$ combined 0.7463). R1b extended this to per-day circadian curves over each participant's first 40 adequate days on the R1a-AF folds: day-level AUROC rises by $+0.0193$ (coverage-residualized: $+0.0181$); participant-level aggregation preserves a $+0.010$ increment over the R1a-AF baseline. All three increments are Bonferroni-positive. The day-scale experiment traces the prospective question: how does performance scale with the number of days sampled per participant? It rises from 0.6653 at $k=1$ to 0.7269 at $k=32$ and 0.7302 with all days; personal-baseline features help most when k is small.

4 Temporal representations (main result)

Question: does *within-day* structure, beyond what the daily summaries already capture, add information? We bin each participant's first 40 adequate Garmin days into 288 five-minute bins (local time; $5 \text{ min} \times 288 = 24 \text{ h}$), bin-mean the heart rate per user-day, mask unobserved bins, and learn several representations of the resulting matrix. The head is ridge regression with α chosen on validation (alloc 0):

$$\hat{w} = \operatorname{argmin}_w \|y - Xw\|_2^2 + \alpha \|w\|_2^2 \quad (3)$$

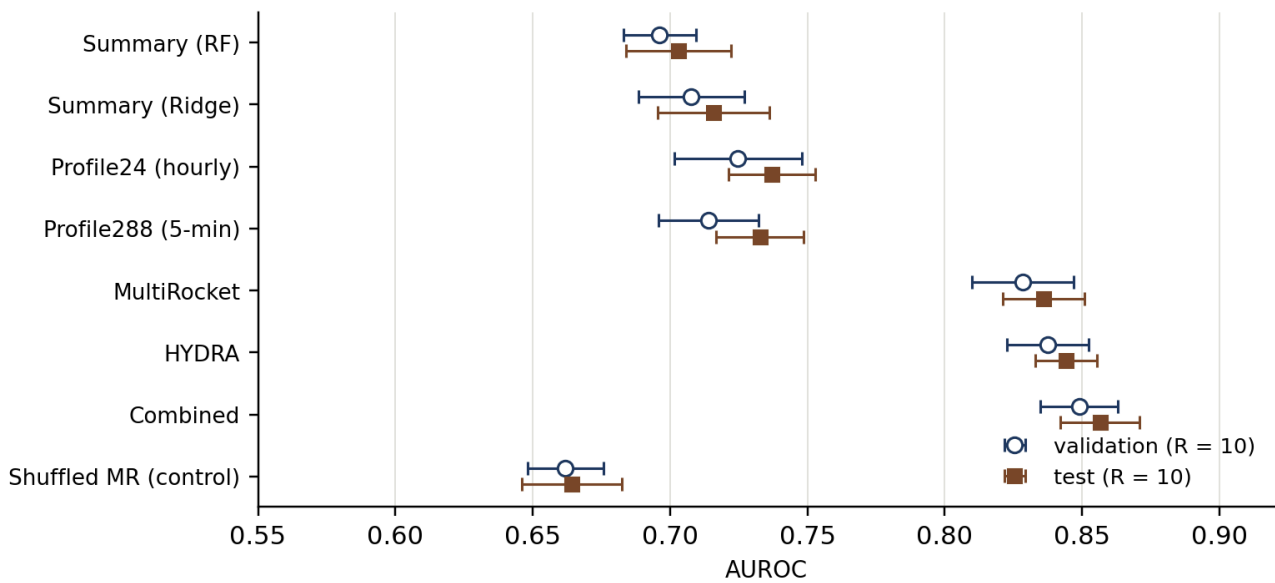
The two convolutional kernels used here are MultiRocket and HYDRA. Both apply a dilated 1-D convolution across each day's series:

$$(k * x)[t] = \sum_j w_j \cdot x[t + d \cdot j] \quad (4)$$

with kernel length 9 and dilations $d \in \{1, 2, 4, 8, 16, 32\}$; the per-day feature vector is max- and min-pooled over a small set of *competing* kernels and pooled across the 40 days by nanmean/nanstd across observed days. The Combined arm concatenates the summary, MultiRocket and HYDRA blocks; Summary_RF is the random-forest control on the summary block alone.

Table. Temporal representation ladder on the R1b cohort/folds.

Representation	validation AUROC (mean $\pm$ SD, R=10)	test AUROC (mean $\pm$ SD, R=10)	post-hygiene cols	α (frozen)
Summary (RF)	0.6962 $\pm$ 0.0132	0.7031 $\pm$ 0.0192	305	—
Summary (Ridge)	0.7077 $\pm$ 0.0193	0.7157 $\pm$ 0.0204	305	1000
Profile24 (hourly)	0.7248 $\pm$ 0.0233	0.7370 $\pm$ 0.0157	353	1000
Profile288 (5-min)	0.7140 $\pm$ 0.0181	0.7327 $\pm$ 0.0159	881	1000
MultiRocket	0.8284 $\pm$ 0.0184	0.8361 $\pm$ 0.0149	19121	1000
HYDRA	0.8374 $\pm$ 0.0149	0.8442 $\pm$ 0.0113	12593	1000
Combined	0.8489 $\pm$ 0.0141	0.8565 $\pm$ 0.0146	31409	1000
Shuffled MR (control)	0.6619 $\pm$ 0.0139	0.6643 $\pm$ 0.0183	19121	1000

**Figure 1.** Same data as Table 2. Error bars are ± 1 SD across 10 allocations. The dashed line marks uninformative ranking (AUROC = 0.5).

4.1 Primary paired deltas

Table. Paired deltas vs Summary_RF on validation (Bonferroni two-sided, m=4, df=9). The last row isolates the placement signal within MultiRocket.

Comparison	mean Δ	SD	Bonferroni 95%-simult	share > 0
Profile288 (5-min) – Summary (RF)	+0.0177	0.0135	[+0.0045, +0.0310]	0.90
MultiRocket – Summary (RF)	+0.1322	0.0241	[+0.1085, +0.1558]	1.00
HYDRA – Summary (RF)	+0.1412	0.0208	[+0.1207, +0.1617]	1.00
Combined – Summary (RF)	+0.1527	0.0207	[+0.1323, +0.1730]	1.00
MultiRocket – Shuffled MR (placement isolation)	+0.1664	0.0225	[+0.1443, +0.1886]	1.00

The combined model adds **+0.1527** over Summary_RF (paired, allocs-positive); the within-day placement isolation (MultiRocket – Shuffled MR) is **+0.1664**. The within-day value-shuffle control permutes observed heart-rate values among each day's observed clock-bins, leaving the value multiset, the wear mask, and the day count intact. The collapse from 0.8284 to 0.6619 says the gain is *where in the day values sit*, not which values are present.

4.2 What the result does and does not say

Profile24 (hourly aggregation) and Profile288 (5-minute aggregation) reach only the summary level, so resolution is not the bottleneck — representation is. Per-bin daily profiles discard the local dilation/position patterns the convolutional kernels exploit at the same 5-minute resolution. We do not establish whether the placement signal is physiological (circadian phase and shape) or device-behavioral (wear-time routines correlated with the recorded salutation); the predeclared probes are night-only arms, activity-window exclusion, and importance-by-dilation.

5 Selective classification

AUROC measures ranking; the applied goal is to *label* a subset of participants at a target correctness and abstain on the rest. Per allocation, validation scores pick the smallest class-1 threshold achieving at least the target precision on validation, and the largest class-0 threshold doing the same on the negative tail:

$$\tau_1 = \min\{\tau : \hat{P}(y = 20 \mid s \geq \tau) \geq p_{\star}\} \text{ (validation), precision} = s/(s+f), \text{ coverage} = \#\{s \geq \tau\}/n \quad (5)$$

Table 4 reports coverage at 95% and 98% target precision in two variants: `raw` (the empirical threshold) and `cpc` (the largest set whose 90% Clopper–Pearson lower confidence bound on precision still clears the target):

$$L(s, f) = q_{0.10}(\text{Beta}(s, f+1)); \text{ certify } \tau \text{ if } L(s, f) \geq p_{\star} \quad (6)$$

Table. Coverage at per-class target precision. prec columns are the mean over allocations of per-allocation test precision (class 20 / class 10). cov_total is the fraction of the test cohort labeled as either class under the per-class thresholds; unattainable thresholds are excluded.

Arm	target	variant	cov_total (%)	prec (class 20)	prec (class 10)
Combined	95%	raw	25.2 ± 10.5	0.956	0.937
Summary (RF)	95%	raw	2.0 ± 2.1	0.873	0.660
BASE ⊕ P40 (R1b baseline)	95%	raw	12.8 ± 9.1	0.905	0.852
Random (sanity)	95%	raw	0.2 ± 0.3	0.644	0.500
Combined	95%	cpc	8.1 ± 9.2	0.959	—
Summary (RF)	95%	cpc	0.0 ± 0.0	—	—
BASE ⊕ P40 (R1b baseline)	95%	cpc	0.0 ± 0.0	—	—
Random (sanity)	95%	cpc	0.0 ± 0.0	—	—
Combined	98%	raw	12.3 ± 8.9	0.971	0.945
Summary (RF)	98%	raw	1.8 ± 1.6	0.867	0.660
BASE ⊕ P40 (R1b baseline)	98%	raw	7.8 ± 6.1	0.930	0.872
Random (sanity)	98%	raw	0.2 ± 0.3	0.644	0.500
Combined	98%	cpc	0.0 ± 0.0	—	—
Summary (RF)	98%	cpc	0.0 ± 0.0	—	—
BASE ⊕ P40 (R1b baseline)	98%	cpc	0.0 ± 0.0	—	—
Random (sanity)	98%	cpc	0.0 ± 0.0	—	—

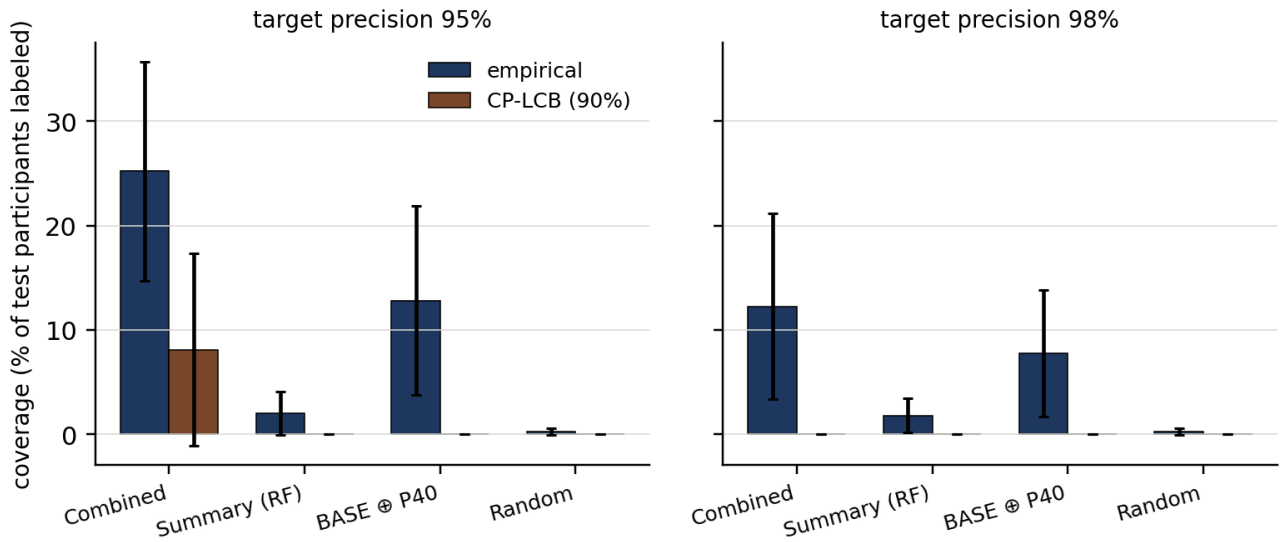


Figure 2. Same data as Table 4 (cov_total panel only). Left: target = 95%. Right: target = 98%. CP-LCB is the 90% lower bound on precision (conservative; rejects sets whose calibration-set precision is uncertain).

At 95% target, the Combined model labels 25.2% of held-out participants (per-class precision 0.956 / 0.937), versus 12.8% for the R1b BASE⊕P40 baseline (ratio $\approx \times 1.97$). The CP-LCB variant certifies only 8.1% at 95% and nothing at 98% on the Combined arm at the available validation size ($n=579$); reaching 98% under finite-sample certification requires roughly 114 error-free validation selections at $\delta=0.10$, which the model cannot attain.

Two qualifications. First, the threshold sits on the validation precision boundary, so held-out precision shrinks with selected-set size — visible on the minority class (class 10), where selected sets are small and precision drops mildly below the target. Second, abstention composition shows that abstained and labeled participants differ by mean heart rate (~ 71 vs ~ 74 bpm) and not by wear coverage (~ 0.90 vs ~ 0.88 mask coverage; ~ 261 vs ~ 254 observed bins/day): the separator is physiological ambiguity, not data scarcity, so minimum-wear gating is not the binding lever for higher coverage.

6 Regularization headroom

The ridge head (Eq. 3) was precalibrated on a grid $[10^{-3}, \dots, 10^3]$; every family in the v1 and v2 runs selected the grid maximum $\alpha=10^3$ (REPORT §8 caveat). An extended-grid addendum on allocation 0 sweeps $[10^2, 3 \cdot 10^2, 10^3, \dots, 10^6]$:

Table. Extended- α addendum (allocation 0; predeclared procedure, NEXT_STEPS §Step 0.7).

Family	frozen α	val @ frozen	α^*	val @ α^*	gain (val)	test @ frozen	test @ α^*
Summary_linear	1000	0.7161	300	0.7176	+0.0015	0.7107	0.7118
Profile24	1000	0.7459	1000	0.7459	+0.0000	0.7404	0.7404
Profile288	1000	0.7204	10000	0.7343	+0.0138	0.7339	0.7178
MultiRocket	1000	0.8234	3000	0.8348	+0.0114	0.8507	0.8626
HYDRA	1000	0.8194	10000	0.8468	+0.0274	0.8401	0.8674
Combined	1000	0.8446	10000	0.8598	+0.0152	0.8625	0.8784

The convolutional arms find optima well inside the extended grid (MultiRocket $\alpha^*=3 \cdot 10^3$; HYDRA and Combined $\alpha^*=10^4$), and the val gains are test-corroborated at this allocation (Combined +0.0159 test).

Profile288 is the exception: it shows +0.0138 on validation at $\alpha^*=10^4$ but -0.016 on test — the alloc-0 val- α

selection-overfit signature, not carried forward. The frozen decision rule (Combined val gain ≥ 0.01) was triggered, and a v3 full rerun is pending the user's decision. The wider lesson: for very wide feature matrices (Combined $\approx 31,000$ columns after hygiene), the v2 grid truncates the optimum and the headline AUROC is if anything understated.

7 Limitations and threats to validity

1. **Label noise ceiling.** If a fraction p of recorded salutations are effectively arbitrary (mis-entered, transiently changed), the maximum achievable AUROC against the recorded label is bounded by

$$\max \text{AUROC} \approx 1 - p/2 \quad (7)$$

2. **Cohort selection.** The R1b cohort is coverage-gated (Garmin users with 40 adequate heart-rate days). Absolute AUROCs are not estimates of performance in the full Datenspende population, nor among participants with missing salutation. The informative-recording experiment does not identify an MNAR mechanism.
3. **Reused test split.** The recent experiments reuse the R1a-AF allocator folds; the test participants have already supported R1a-AF and R1b model development and are reported as exploratory by predeclared rule.
4. **Day-scale sampling caveat.** The day-scale k-curve samples days *with replacement* across each participant's full record; the $k=1$ point is therefore not a one-day-of-history number. R1b's chronological first-k curve addresses the prospective question on the R1b model (first day 0.669, near-plateau by day 32).
5. **Threshold guarantees are conditional.** The selective-classification raw numbers are operating estimates; the CP-LCB certification is finite-sample on the validation split (conditional on the split), not a population promise. With $n=579$ and $\delta=0.10$, certification at 98% precision is unattainable.
6. **v1 $\leftrightarrow$ v2 robustness.** The v1 temporal run had a zero-sentinel contamination defect (unobserved bins stored as 0.0 and included by Profile288 and the per-bin fill). Table 6 quantifies the effect: every arm moved by ≤ 0.004 AUROC between v1 and v2, and the B40-only arms are bit-identical. v1 is retained as the erratum record (commit 168b9d7); v2 is primary.
7. **Allocation variability is not population uncertainty.** Reported intervals describe model-and-split randomization on the observed cohort.

Table. v1 (erratum present) vs v2 (mask-aware fix). B40-only arms are bit-identical.

Representation	val v1	val v2	Δ val	test v1	test v2	Δ test
Summary (RF)	0.6962	0.6962	+0.0000	0.7031	0.7031	+0.0000
Summary (Ridge)	0.7077	0.7077	+0.0000	0.7157	0.7157	+0.0000
Profile24 (hourly)	0.7248	0.7248	+0.0000	0.7370	0.7370	+0.0000
Profile288 (5-min)	0.7165	0.7140	-0.0026	0.7291	0.7327	+0.0036
MultiRocket	0.8276	0.8284	+0.0008	0.8354	0.8361	+0.0007
HYDRA	0.8392	0.8374	-0.0018	0.8431	0.8442	+0.0010
Combined	0.8496	0.8489	-0.0007	0.8539	0.8565	+0.0026
Shuffled MR (control)	0.6656	0.6619	-0.0036	0.6657	0.6643	-0.0014

8 Reproducibility record

Every stage was run from a frozen plan committed before any results were seen; deviations from a frozen plan are recorded as errata in the per-stage REPORT, never silent. Determinism is gated on each run: a full alloc-o

rerun must be byte-identical or differ by at most 10^{-6} per arm. The actual rerun differences are 1–2 unit-last-place (max $2.2 \cdot 10^{-16}$), attributable to threaded-BLAS reduction order in the ridge solve (the same signature is present in pure-B40 Summary_linear, so it is not a transform artifact). MR and HYDRA transforms, and all seed streams, are bit-stable.

Resource gates: CPU torch only, peak resident-memory under 8 GiB across all stages (v2 plateau 8,197 MiB, within +0.06% of the gate; addendum peak 6,309 MiB). Vendored MultiRocket (commit 3ccaa4f) and HYDRA (commit 144bb7a) are pinned with file SHA256s in cache/vendor_repro.json. A numba-seeding shim (_nb_seed) and the documented workqueue threading layer are the two recorded environment errata.

9 Next steps

Three follow-ups are plan-ready and pending review or decision:

1. **v3 full rerun** with the extended α -grid from alloc-o precalibration (72 min; no other changes). Expected Combined test AUROC ≈ 0.87 – 0.88 ; selective-classification numbers carry forward under the same predeclared procedure.
2. **Multichannel plan** (draft at NEXT_STEPS §5 Appendix A): Steps, MET, walk/activity, sleep binaries, resting-HR added to the same 288-bin $\times$ 40-day grid; multichannel HYDRA (native) and per-channel MultiRocket. Co-primary: AUROC ≥ 0.90 and raw coverage@95% $\geq 35\%$ (vs Combined 25.2%) and CP-LCB $\geq 15\%$ (vs 8.1%).
3. **Artifact-discrimination probes**: night-only transform, activity-window exclusion, importance-by-dilation — to separate physiological placement from device-behavioral placement. Fold into the multichannel experiment as sensitivity arms.

None of the above establishes a causal or physiological claim; the documented scope is predictability from the export.

A Notation and file index

Notation. s_i model score for participant i ; τ threshold; $p_\star$ target precision; α ridge penalty; d convolution dilation; R number of allocations (here 10); m number of primary comparisons in the Bonferroni family; $L(s, f)$ Clopper–Pearson 90% lower bound on the success probability of a binomial with s successes and f failures; δ miscoverage target (0.10 here).

Files. experiments/temporal_garmin/results/REPORT.md (full per-arm results, §1–§10); experiments/temporal_garmin/results/temporal_metrics.csv and temporal_metrics_v1.csv (per-arm AUROCs); experiments/temporal_garmin/results/selective_classification.{csv,md} (per-alloc coverage-at-precision); experiments/temporal_garmin/results/alpha_addendum.{json,md} (extended- α grid); experiments/temporal_garmin/results/risk_coverage_curves.csv (full operating curves); experiments/temporal_garmin/results/abstention_composition.csv (labeled vs abstained participant statistics); experiments/temporal_garmin/NEXT_STEPS.md (roadmap + multichannel draft + selective-classification archived procedure).